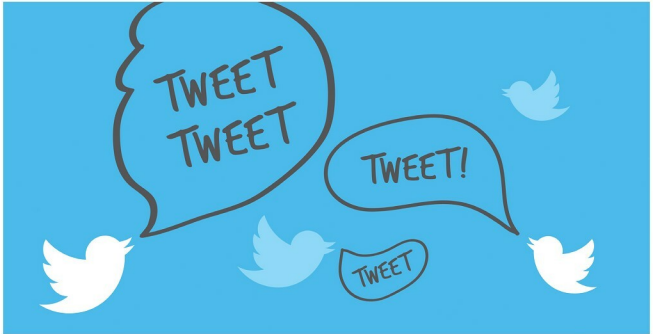


Using Python to Automate Retweets

"By 2022, most Platform-as-a-Service (PaaS) offerings will evolve to a fundamentally serverless model, rendering the cloud platform architectures that are dominant in 2017 as legacy architectures." - Gartner



In the September 2017 issue of *OSFY*, we demystified serverless computing. We also discussed its pros and cons, the technologies involved, use cases, the challenges, etc. So how about using serverless computing to simplify or automate your tasks? For example, wouldn't it be nice to automatically retweet the tweets that have your favourite hashtags (such as `#serverless` or `#opensource`). That's the topic we'll explore in this article. We are going to implement a bot (let's call it *TweetBot*), in which you can specify the hashtags to retweet at specified time intervals.

We will start with writing an auto-retweeting code in Python and get it working on our machine. Later, we will deploy it in OpenWhisk on IBM Bluemix. For this article, we assume you already know the fundamentals of Python programming. You need not know anything about OpenWhisk or Bluemix — we'll introduce these to you in this article.

Developing the TweetBot

In order to write the Python code for auto-retweeting, the prerequisite is that Python 3 must be installed in your machine.

The Twitter API provides programmatic access to read and write Twitter data, create a new tweet, read user profile and follower data, retweet, and more. For this, you must first create a Twitter application (see <https://apps.twitter.com/>) and note down the OAuth credentials for the bot configuration.

The Twitter API in Python can be accessed using the *TwitterFollowBot* Python module. Using this module, you can do much more than just retweet — you can auto-follow and auto-like. For developing the *TweetBot*, you must install the module into a folder rather than the default *bin* directory. For that, use the following command:

```
pip install --target <target_folder> TwitterFollowBot
```

Let us now create a program that uses the *TwitterFollowBot* Python module to retweet the latest 'count' number of tweets with a specific phrase ('*#Serverless*' in our case). For that, we need to create a *TwitterFollowBot* instance. The *TwitterFollowBot* uses the *config.txt* file in the current directory to configure the bot.

So, let us first configure the bot. To do so, you should create a *config.txt* file and fill in the following information so that the bot can connect to the Twitter API. Here are the entries for the *config.txt* file:

```

AUTH_TOKEN:
AUTH_SECRET:      API keys from
CONSUMER_KEY:      the twitter app
CONSUMER_SECRET:
TWITTER_HANDLE: <your twitter handle>
ALREADY_FOLLOWED_FILE: already-followed.txt
  
```